

Deep Learning to mitigate Economic Denial of Sustainability (EDoS) Attacks: Cloud Computing

1st Husam Ibrahim Husain Alsaadi

Computer Science Department,
College of Basic Education,
Mustansiriyah University,
Ministry of Higher Education and Scientific Research,
Baghdad, Iraq.
husam.alsaadi@uomustansiriyah.edu.iq

2nd Maad Kamal Al-anni

Computer Engineering department,
College of Engineering,
Al-Iraqia University,
Ministry of Higher Education and Scientific Research,
Baghdad, Iraq.
muad.rashed@aliraqia.edu.iq

3th Fanar Emad Khazaal Al-Khuzai

Babylon Education Directorate,
Ministry of Education,
Babylon, Iraq.
fanar@itnet.uobabylon.edu.iq

Abstract—Cloud computing, despite being a groundbreaking technology, has increased the risk of attacks that exploit cloud services, leading to the exhaustion of resource allocations. One such attack is Economic Denial of Sustainability (EDoS), which uses pay-per-use services to gradually increase resource demand, potentially leading to financial disaster for service provider.

The Cyber Range Lab at the University of New South Wales (UNSW), located in Canberra, conducted nine injection attacks to test machine learning and deep learning algorithms. These experiments involved multi-classification, incorporating nine different types of attack data. Among the presented algorithms, the RNN-LSTM method achieved an accuracy of 84%. Various statistical analyses, such as mean square error (MSE), Pearson correlation coefficient (R), and root mean square error (RMSE), were used to evaluate prediction errors between input data and values generated by different machine learning and deep learning algorithms. Despite its relatively low prediction level (MSE=0.560), the Recurrent Neural Network Long Short Term Memory (RNNLSTM) algorithm achieved an R2 level of 82.09% when applied to the multi-classification dataset. The suggested system's performance was compared to existing EDoS attack detection systems, and the RNN-LSTM-based attack mitigation algorithms demonstrated superior performance. This study aims to contribute to the identification and effective protection of cloud-based resources that could be exploited by Distributed Denial-of-Service (DDoS) attacks, thereby preventing financial losses for potential victims.

Index Terms—Deep Learning, Cloud Computing, Economic denial of sustainability (EDoS), Artificial Intelligent, Intrusion Detection System (IDS), and Denial of Service (DoS).

I. INTRODUCTION

By overwhelming the target systems with traffic, Denial of Service (DoS) assaults hope to render them inoperable. Single-source threats are commonly referred to as DoS, whereas multi-source threats are referred to as Distributed Denial-of-Service assaults (DDoS). These dangers have

evolved over the last several decades to become more pervasive, more sophisticated, and harder to counteract.

Organizations specializing in data protection have issued warnings about this same issue. For instance, DDoS attacks have increased by 30% over the last year, as reported by the European Union Agency for Network and Information Security (ENISA) [1]. Given the scope of the issues reported in the fall of 2017 [2], the European Commission (EC) [3] [4] and the United States Government [5] have both proposed significant new steps to counteract these threats.

The European Police (Europol) claims that several factors contribute to the growing threat posed by cybercriminals, including the exponential growth of botnets, the appearance of new vulnerabilities and amplifying elements, the increased availability of malicious products as Crimeware-as-a-Service (CaaS) on the dark web, the widespread adoption of certain technologies (e.g., mobile devices, the Internet of Things [IoT]), and users' lack of knowledge about proper cyber hygiene.

In the context of the economy, an attack known as a denial-of-service (DoS) may also be referred to as an economic denial of sustainability (EDoS) [6]. EDoS attacks are a waste of resources because they force unnecessary scaling of services by taking advantage of the pay-per-use paradigm offered by cloud computing. When compared to conventional DoS attacks, particularly those that are based on flooding [7], the volume of traffic that is generated during an EDoS attack is often smaller.

Therefore, distributed denial of service attacks, also known as EDoS, are a kind of low-rate distributed denial of service attacks that are difficult to detect using methods that are dependant on the volume of traffic. The EDoS is produced by bombarding victims with tens, hundreds, or even millions

of individual attack packets in the space of a single second. Consequently, victims would experience acute exhaustion even if the condition was discovered one second later. As a consequence of this, the capability of the system to swiftly spot abnormalities is essential.

In addition, one purpose of denial of service attacks is to bring the targeted service provider to its knees and force it to shut down. Because of this, they are able to remain hidden for a considerable amount of time while simultaneously draining the cloud's resources and deceiving the majority of threshold-based DoS security solutions. The structure of EDOS attack is displayed in Figure.1.

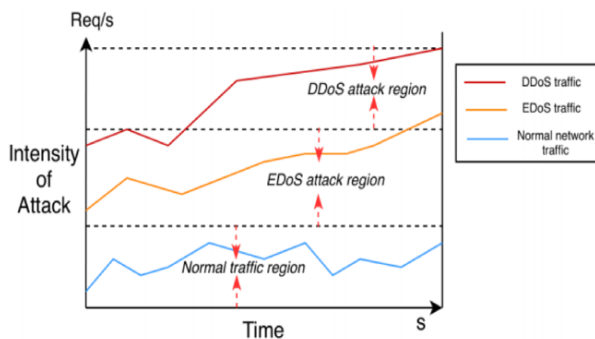


Fig. 1: structure of EDOS attack.

There have been several strategies [8] described to identify distributed denial of service attacks. It has been shown that machine learning (ML) techniques such as support vector machine (SVM) and self-organizing map (SOM) are successful when it comes to identifying distributed denial of service attacks [9] [10]. When it comes to dealing with the massive amounts of EDoS data that inevitably occur in network application settings, traditional machine learning methods place a significant emphasis on feature engineering and selection.

This, however, poses complications. Deep learning (DL), which is a subset of machine learning, is automatically utilized to better extract features by applying neural layers. This not only improves the accuracy and durability of the detection technique, but it also overcomes the limitations of ML. The structure of the human nervous system serves as an inspiration for the many neural network models that are employed as classification algorithms in DL. These algorithms make use of a large number of non-linear processing units to find solutions to challenging problems.

Because EDoS is stealthy, many models and methods that apply flow-based detection techniques need a time-series model that can remember the most recent input and anticipate the output of the sequence data. This is because the time-series model must be able to recall the most recent input. The recurrent neural network (RNN), which is a model for the sequence input, may be used to identify denial of service (DoS) attacks [11]. Long short-term memory (LSTM) models

[12] are another RNN scheme that can tackle the vanishing gradient issue of RNNs. These models have an internal mechanism called "memory gates" that can govern the flow of the input sequence. This allows the models to solve the problem of RNNs' vanishing gradients. In comparison to previous recurrent models [6] [13] [14], this one takes much longer to train and detect, and it also has memory restrictions. Stacking the memory cells leads the model to lose information about the prior input and quickly vanishes when the back propagation gradient is calculated. This is due to the fact that LSTM calculates the inputs in a progressive manner over the course of time. In addition, when the input is long, the time lag in the training of the model becomes much more evident.

Hoff first identified Economic Denial of Sustainability (EDoS) assaults in 2008 [15] [16], and since then, the term's meaning has been expanded by Cohen [17]. Most distributed denial of service attacks (DDoS) target Cloud Computing networks because of their importance to the development of new forms of communication. According to Singh et al. [18], "threats whose goal is to make the costing model unsustainable and consequently making it no longer possible for a corporation to affordably utilize or pay for its Cloud-based infrastructure" are the official definition of EDoS assaults. Bibliographic references on EDoS also use the terms "RoQ threats" [19] and "FRC assaults" [20] to describe similar types of attacks that include the fraudulent use of resources. To accomplish their goals, these attacks use the "pay as you go" billing structure adopted by the majority of Cloud Computing providers and their auto-scaling services [19]. Their mode of operation [21] and the scaling strategies they enact differ differently depending on the providers and the Cloud solutions they provide (such as OpenStack, Microsoft Azure, Amazon EC2, etc.). (discrete, adaptive, etc.). When an EDoS attack occurs, the attacker often injects requests that the server must handle. Large file or query requests [22], HTTP requests on XML files [22], and exploiting alternative Application layer vulnerabilities [23]–[27] are just a few examples of the ways in which these threats might increase the strain. Auto-scaling mechanisms kick in when the volume of requests overwhelms the computing capacity of the contracted services, resulting in an increase in the client's bill.

The main contribution of the proposed research is to address the limitations of machine learning when applied to the EDoS detection problem, we propose a new RNN-LSTM approach that makes advantage of parallel computing and persistent storage to expedite the training and prediction stages. In this part, we will provide a high-level overview of the most important findings. Here, we provide an effective method for detecting EDoS by using the attention technique.

II. RELATED WORKS

Mohamed Idhammad et al. introduce a distributed Machine Learning-based intrusion detection system for Cloud environments [28], addressing security issues in cloud computing. Utilizing a time-based sliding window algorithm and ensemble learning classifiers, the system, when tested on the CIDDS-001 dataset in Google Cloud Platform, achieved 97% accuracy with minimal false positives.

Bakro et al. proposes a cloud intrusion detection system, enhancing classification accuracy through improved feature selection and an ensemble model weighted with the crow search algorithm [29]. Experiments on multiple datasets show superior performance in accuracy, recall, and precision compared to traditional methods. The ensemble model demonstrated robustness against false positives and negatives.

Al-anni et al. introduce a Genetic Algorithm approach for misuse detection in programs [30]. It outlines Intrusion Detection Systems and related techniques. While manual rule development is significant, defining thresholds is challenging. Thus, we propose learning intrusion detection rules using genetic algorithms. Effectiveness is tested on KDD cup 99 and UoP datasets, focusing on specific attacks like Smurf and Apache2.

Al-anni explores the Artificial Neural Network's reliance on Multilayer Perceptron (MPL) for classifying and clustering Mobile Adhoc Network vulnerabilities [31]. The mobile ad hoc network (MANET) comprises intelligent devices that autonomously detect their environment. Given the security concerns in MANET, this paper emphasizes the effectiveness of artificial neural networks in intrusion detection. By simulating node attacks and contrasting with normal behaviors, any deviation is flagged as an attack. This paper showcases a signature-based IDS in MANET using the back propagation algorithm, achieving a 98.6% detection rate. Performance metrics, including Packet Delivery Ratio and Throughput, are discussed.

Parampottupadam et al. examine real-time network intrusion detection using deep learning [32]. A cloud prototype system was designed and evaluated with the NSL-KDD dataset, comparing deep learning libraries against traditional machine learning models. Results indicated that the choice of deep learning library significantly impacts real-time performance, with H2O models excelling in accuracy.

Rafah M. Almuttairi et al. present their research that the Promotion and recognition can be achieved by utilizing web applications and electronic media to showcase a company's offerings. Protecting websites from attacks [33], especially Distributed Denial of Service (DDoS) attacks at the application layer, is crucial for maintaining reputation. Traditional detection systems struggle due to increasing attack data volume. This study proposes a method leveraging the Hadoop framework for swift data analysis and detection. Detected attacks are blocked using Snort. Results showed significant reduction in alerts compared to the traditional model, with reduction rates of 99.01%, 99.27%, and 99.72% for specific alerts.

Alsaadi, H. I. et al. introduce computational intelligence models, specifically Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO) [34], for feature selection in intrusion detection systems. These methods optimize classifier algorithms, enhance processing times, and outperform conventional models using benchmark datasets like KDD cup and NSL-KDD.

Alanni highlights security concerns in Grid computing by a protection shield [35]. Grid Security Infrastructure (GSI) protects against external attacks using certificates for user authentication. Despite security measures, internal attacks remain a concern, particularly during delegation processes in middleware software like the Globus toolkit. The study proposes using Neural Networks with Intrusion Detection Systems to enhance security by identifying abnormal patterns and behaviors.

Alsaadi, H. I. et al. introduce computational intelligence models [36], specifically ACO and PSO, for feature selection in intrusion detection systems. These models optimize classification algorithms like KNN, SVM, and NB. Using KDD and NSL-KDD datasets, the proposed model demonstrated superior performance over conventional models in intrusion detection with 89.98% in Dos attack.

F. E. K. Al-Khuzaie et. al use deep learning to diagnose Alzheimer's from 2D MRI brain slices [37], differing from previous 3D-focused methods. Named "AlzNet", this model achieved 99.30% accuracy, outperforming prior techniques and offering promising diagnostic potential.

III. MATERIALS AND METHODS

In this section, the developed system's framework for detecting EDoS attacks on the cloud computing environment is presented in Figure.2 .

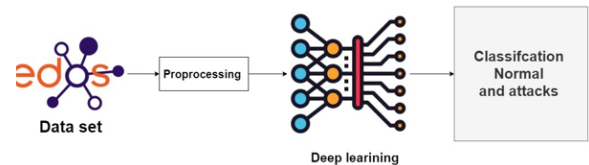


Fig. 2: Framwork of deep learning.

A. Datasets

The IXIA PerfectStorm application was used in a Cyber Range Lab at the University of New South Wales (UNSW), located in Canberra, to replicate both real-world, daily cyber activity as well as complex, targeted cyber attacks. Following the collection and subsequent analysis of raw data totaling one hundred terabytes using the tcpdump application (e.g., Pcap files). The dataset contains various different types of injection attacks, such as Fuzzers, Analyzers, Backdoors, Denial of Service Attacks, Exploits, Generic Reconnaissance, Shellcode, Backdoors, and Worms. Figure.3 demonstrates that a total of Argus, Bro-IDS, and 12 distinct algorithms were used in order to produce the 49 features that were found. .

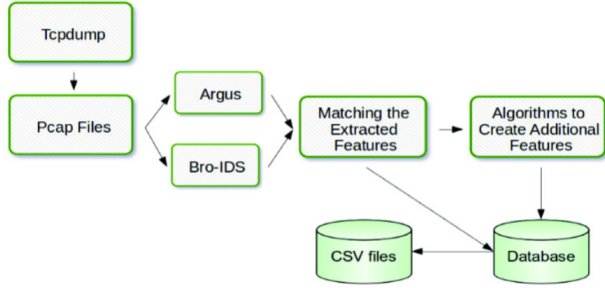


Fig. 3: Framework architecture and workflow of generating the UNSW-NB15 data set.

B. Preprocessing

There were a total of 49 characteristics and 175333 rows in the dataset. Following the elimination of missing data, the dataset had 45 characteristics and 80972 rows. Attribute data type changes are made using the data type information provided by the features dataset.

1) *One-hot encoding*: The one-hot encoding approach was used to turn categorical features, such as "protocol," "service," and "state," into numerical values, which assisted in categorization and helped identify assaults.

C. Feature Selection Algorithm

Principal Component Analysis (PCA) is a dimensionality reduction technique used for feature selection. It transforms the original features into a new set of uncorrelated variables (principal components) while retaining the most important information. It helps eliminate redundant and less informative features, enhancing the efficiency and performance of machine learning algorithms.

1) *Min-max normalization method*: Normalization makes use of the min-max methodology, which is one of the most popular methods of doing so. Each feature's significance is ranked from least to greatest; the least important value is assigned the value 0, the most significant value is assigned the value 1, and all other values are converted to decimals between 0 and 1. When using the min-max normalization technique, the following equation is utilized:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

D. Recurrent neural network (RNN)

The recurrent neural network (RNN) is considered to be the gold standard for training temporal data; nevertheless, traditional RNNs are notoriously difficult to train due to gradient explosion or disappearance. The LSTM algorithm solves these problems by replacing the RNN's hidden units with units that have a memory function built into them. LSTM is capable of long-term memory and can also trigger short-term memory in a range-limited manner due to the fact that its weight varies very little over time. The main data stream of the LSTM moves down the horizontal axis, and the forget gate, input gate, and output gate give it the ability to forget the data it has already processed and take in fresh

information.

Figure.4 shows that the current state may be calculated by combining the word vector W_t with the previous hidden state h_{t-1} and the previous cell state c_{t-1} . Since this is the case, the following equation holds:

$$f_t = \sigma(W_f.X_t + W_f.h_{t-1} + b_f) \quad (2)$$

$$i_t = \sigma(W_i.X_t + W_i.h_{t-1} + b_i) \quad (3)$$

$$S_t = \tanh(W_c.X_t + W_c.h_{t-1} + b_c) \quad (4)$$

$$C_t = (i_t.S_t + f_t * C_{t-1}) \quad (5)$$

$$o_t = \sigma(W_o.X_t + W_o.h_{t-1} + V_o.C_t + b_o) \quad (6)$$

$$h_t = o_t + \tanh C_t \quad (7)$$

using both the tanh function, which is short for hyperbolic

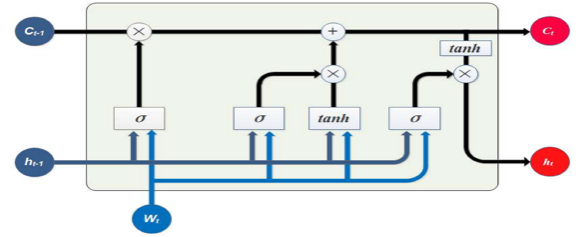


Fig. 4: structure of RNN-LSTM .

tangent, and the sigmoid function. In mathematical notation, the sign stands for matrix multiplication, while the symbol denotes element-wise multiplication. The input gate I, the forget gate (f), and the output gate (o) all use weight matrices (Ws) and hidden state vectors to determine whether or not the state of the cell is updated with the input, whether or not the memory from the previous time step is forgotten, and whether or not the memory is output. This is done in order to determine whether or not the memory is output (Us).

E. Performance measurements

The success of the proposed algorithms in recognizing EDoS malwares is evaluated using a variety of measures, such as the Sensitivity, Specificity, Precision, Recall, F1 Score, Mean Square Error (MSE), Pearson's Correlation Coefficient (R), and Root-Mean-Square Error (RMSE). Parameter equations are as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_{i,exp} - y_{i,pred})^2 \quad (8)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_{i,exp} - y_{i,pred})^2}{n}} \quad (9)$$

$$R^2 b_{n1} = \frac{\sum_{i=1}^n (y_{i,exp} - y_{i,pred})^2}{\sum_{i=1}^n (y_{i,exp} - y_{avg,exp})^2} \quad (10)$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \times 100\% \quad (11)$$

$$Sensitivity = \frac{TP}{TP + FN} \times 100\% \quad (12)$$

$$Precision = \frac{TP}{TP + FP} \times 100\% \quad (13)$$

$$Fscore = \frac{2 * precision * sensitivity}{precision + sensitivity} \times 100\% \quad (14)$$

$$R\% = \frac{n(\sum_{i=1}^n y_{i,exp} \times y_{i,pred}) - (\sum_{i=1}^n y_{i,exp})(\sum_{i=1}^n y_{i,pred})}{\sqrt{[n(\sum_{i=1}^n y_{i,exp})^2 - (\sum_{i=1}^n y_{i,exp})^2][n(\sum_{i=1}^n y_{i,pred})^2 - (\sum_{i=1}^n y_{i,pred})^2]}} \times 100\% \quad (15)$$

When the input data is in the form of cloud computing network data (y (y(i,exp))), the projected value of the data point I is represented as(y (y(avg,exp))). In the notation used to express the Pearson correlation coefficient, (y (y(i,exp))), y I (y(i,class)), where I is the number of input data points and n is the number of network data classes, is the set of all possible values for i. yavg,pred, and R are the symbols for summing up the experimental values. Pearson's coefficient of correlation is written as y =y(i,exp). There are four potential results: a true positive (TP), a true negative (TN), a false positive (FP), and a false negative (FN).

IV. EXPERIMENTAL ANALYSIS

In this part, we provide the classification performance of the RNN-LSTM techniques according to assessment criteria such as accuracy, precision, recall, f-score, MSE, RMSE, and R2. The classification algorithms were analyzed on two levels: binary and multi-classification. The binary classification considers two classes, normal or attacks, while the multi-classification includes nine classes: Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic Reconnaissance, Backdoor, Worms, and normal detection. These classifications aim to enhance the cloud computing environment's ability to defend itself against any threat from these attacks.

To accomplish their respective 49 feature sets were used. As a result of employing correlation algorithms, we were able to zero down on the best 49 traits. Robustness between characteristics and the considered class label exists. Results from the suggested algorithms were compared to those from a wide range of already available systems. Following this, we'll talk about the extensive experiment that was conducted for this study.

Tensor Flow and Keras were used to construct deep learning methods such as CNN and LSTM models, while Scikit-learn was used to produce SVM, KNN, and RF. Scikit-learn was used to develop SVM, KNN, and RF. The performance evaluation was carried out with the following components: an Intel (R) Core (TM) i7-4770 CPU operating at 3.20 GHz, 16 GB of RAM, and a 64-bit version of Windows 10. After it was recognized that CNN and LSTM were susceptible to overfitting, several dropouts were included into the training process in order to combat this issue.

Seventy percent of the data set was put aside for training purposes, while the remaining thirty percent was set aside for

evaluation. In order to assess the efficacy of machine learning and deep learning, their results were put to the test.

A. Results of machine learning

In Table 1, show results of RNN-LSTM model to detect the EDOS , when it comes to detecting distributed denial of service attacks in the cloud. The dataset was divided such that 70% of it would be used for training, and 30% would be used for testing, so that we could evaluate the model's ability to recognize attacks on binary datasets. We discovered that the highest level of accuracy that we could test to while using the RNN-LSTM approach and classification result was 99%. The confusion metrics of the proposed system is presented in Figure.5.

Attacks	Precision%	Recall%	F1-score%
Analysis	43	0.01	0.07
Backdoor	76	99	86
Dos	93	29	44
Exploits	100	99	99
Fuzzers	41	42	40
Generis	99	99	99
Normal	44	0.4	0.07
Normal	83	49	62
accumulative Accuracy	84%		

TABLE I: Results of RNN-LSTM model for detecting EDOS attack

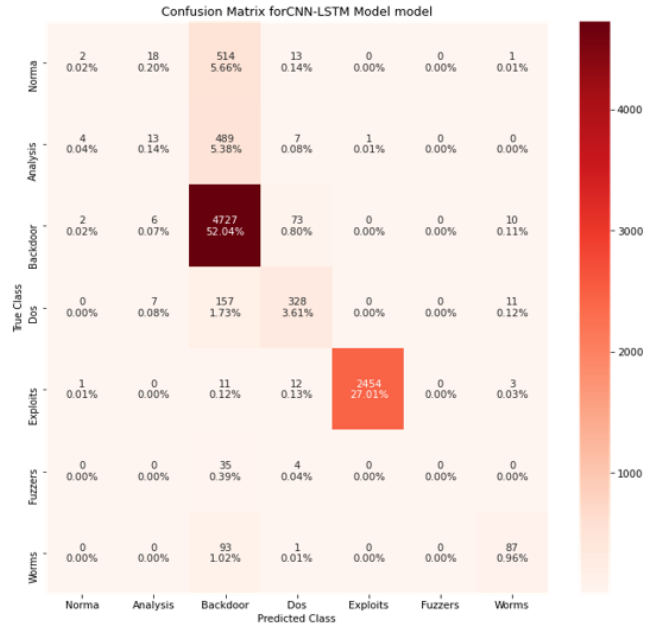


Fig. 5: The visualization displays a confusion matrix for multi-class classification predictions concerning EDoS attacks. The matrix's rows represent observed EDoS attacks, while the columns illustrate the predictions made by the classifier. Each cell contains counts indicating the number of observed attacks. Color gradients, transitioning from white (0%) to dark red (100%), are utilized in each cell to indicate the prediction accuracy for each category. Information about the utilization of the UNSW-NB15 dataset for training and testing can be found above the matrix.

Tests for detecting EDoS assaults on the security system are shown in Figure.6 to demonstrate the proposed model's validation performance. The CNN model started with an accuracy of 83%, and after 20 epochs of training and testing,

it reached a validation accuracy of 84%. Incorporating cross-entropy data allowed for the validation loss to be reduced from its original value of 0.70 to as low as 0.52, which is a significant improvement.

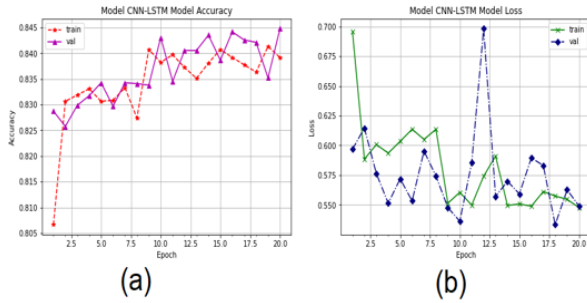


Fig. 6: Performance of RNN-LSTM a)accuracy b)loss

B. Statistical analysis

Data collection and interpretation is what statisticians call "statistical analysis." It's a common step in any study using data. Data collection, result interpretation, statistical modeling, and study design are only few of the many applications of statistical analysis. It's also useful for determining which properties of the dataset and the labels are correlated, as well as for calculating the magnitude of the discrepancy between the predicted and desired values. Statistics for RNN-LSTM model on the dataset are shown in Table II. It is observed that the RNN-LSTM model has achieved $R^2 = 82.09\%$ and Both the mean squared error (MSE) and root mean squared error (RMSE) for RNN-LSTM were very small, at 0.560 and 0.748, respectively.

Models	MAE	MsE	RMSE	R2(%)
RNN-LSTM	0.266	0.560	0.748	82.09

TABLE II: Prediction results o of machine learning and deep learning models using a binary dataset.

table III provides a comparison of the RNN-LSTM model that have been recommended with the systems that are now in place, illustrating how effective the suggested approaches are. Given this high degree of accuracy, we are in a position to declare that the strategy that we have proposed, which is based on multi-classification, is effective. Using RNN-LSTM model resulted in very accurate results as well.

References	Year	Datasets	Classification Type	Model	Feature Selection	Accuracy(%)
Ref. [27]	2019	UNSW-NB15	Multi-classification	SVM	x	75.77%
Ref. [38]	2021	UNSW-NB15	Multi-classification	GA+SVM	x	83%
Ref. [39]	2021	UNSW-NB15	Multi-classification	dynamic classifier	K-best	61%
Ref. [40]	2021	UNSW-NB15	Multi-classification	SVM	x	83.33%
RNN-LSTM	2023	UNSW-NB15	Multi-classification	RNN-LSTM		84

TABLE III: RNN-LSTM Results against existing detection systems.

V. CONCLUSION

Because of the high value of the information they hold, fraudsters often target financial organizations. Sales of stolen login credentials for internet banking have netted cybercriminals millions. As banks expand their online presence, hackers

now have more opportunities to target them. The purpose of our investigation into the UNSW Dataset is to cover information on EDOS attacks against financial institutions and other companies. Algorithms based on deep learning RNN-LSTM has been developed specifically for the goal of identifying financial industry assaults. The RNN-LSTM has achieved a=superior accuracy 84% comparing with different existing systems. Accurate timing predictions are very necessary for cloud computing detection systems due to the fact that EDoS attackers may generate a thousand packets in a very short amount of time. In addition, the amount of time spent training is reduced in half, which is a significant improvement when adding newly found methods of assault by either fine-tuning or training using data that has just been created. This time savings is supposed to be due to the attention mechanism that was predicted. With the assistance of the detection approach that we recommend, EDoS assaults, which have caused financial disasters for a great number of enterprises as a result of their dependence on cloud services, may be averted.

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